

Project: ManuFlow – Production Line Monitoring & Optimization System

1. Introduction

This document outlines the Low-Level Design (LLD) for **ManuFlow**, a production line monitoring and optimization system tailored for manufacturing units. The system digitizes production tracking, monitors machine performance, automates shift planning, and provides analytics for throughput optimization and bottleneck detection.

Supports backend development using **Java (Spring Boot)** and **.NET (ASP.NET Core)**.

2. Module Overview

- 2.1 Production Line Configuration & Shift Planning Module
- 2.2 Real-Time Machine Monitoring Module
- 2.3 Throughput & Bottleneck Analysis Module
- 2.4 Maintenance Alerts & Downtime Logging Module
- 2.5 Operational Analytics & KPI Dashboard Module

3. Architecture Overview

3.1 Architectural Style

- **Frontend:** Angular/React for operator consoles and admin dashboards
- **Backend:** REST APIs for production tracking and analytics
- **Database:** MySQL/SQL Server for structured production and machine data

3.2 Component Interaction

- Frontend captures machine status and shift logs
- Backend processes throughput metrics and generates alerts

4. Module-Wise Design

4.1 Production Line Configuration & Shift Planning Module

4.1.1 Features

- Define production lines and assign machines
- Plan operator shifts and production targets

4.1.2 Entities

- **ProductionLine**
 - LineID
 - Name

- MachinesAssigned
- TargetOutput
- ShiftSchedule

4.2 Real-Time Machine Monitoring Module

4.2.1 Features

- Monitor machine status (RUNNING, IDLE, ERROR)
- Capture operational metrics like speed and temperature

4.2.2 Entities

- **MachineStatus**
 - StatusID
 - MachineID
 - Timestamp
 - Status
 - Speed
 - Temperature

4.3 Throughput & Bottleneck Analysis Module

4.3.1 Features

- Calculate hourly/daily throughput
- Identify bottlenecks in production flow

4.3.2 Entities

- **ThroughputRecord**
 - RecordID
 - LineID
 - Timestamp
 - UnitsProduced
 - BottleneckDetected

4.4 Maintenance Alerts & Downtime Logging Module

4.4.1 Features

- Generate alerts for scheduled and unscheduled maintenance
- Log downtime events and reasons

4.4.2 Entities

- **DowntimeLog**
 - LogID
 - MachineID

- StartTime
- EndTime
- Reason
- MaintenanceType (SCHEDULED, UNSCHEDULED)

4.5 Operational Analytics & KPI Dashboard Module

4.5.1 Features

- Visualize KPIs like OEE, MTTR, and production efficiency
- Export reports for management review

4.5.2 Entities

- **KPIReport**
 - ReportID
 - LineID
 - ReportDate
 - Metrics (OEE, MTTR, Efficiency)
 - Insights

5. Deployment Strategy

5.1 Local Deployment

- Developer machines with simulated machine data

5.2 Staging and Production Environments

- Cloud deployment with secure access and real-time dashboards

6. Database Design

6.1 Tables and Relationships

- ProductionLine → MachineStatus → ThroughputRecord → DowntimeLog → KPIReport

7. User Interface Design

7.1 Wireframes

- **Operator Console:** View machine status, log downtime
- **Admin Dashboard:** Configure lines, plan shifts
- **Analytics Console:** Monitor KPIs and export reports

8. Non-Functional Requirements

8.1 Performance

- Handle 20,000 machine status updates per hour

8.2 Usability

- Touch-friendly UI for factory floor tablets
- Real-time visual indicators for alerts

8.3 Security

- Role-based access and encrypted data streams
- Compliance with ISO 9001 and industry safety standards

8.4 Scalability

- Support for multiple production lines and plants

9. Assumptions and Constraints

9.1 Assumptions

- Machines are equipped with IoT sensors for data capture
- Operators will log downtime via web or mobile interface

9.2 Constraints

- Initial rollout for discrete manufacturing units only